WHO and UNICEF Estimates of National Infant Immunization Coverage

User Reference

Country Reports

01 May 2024

Prepared by the WHO and UNICEF working group for monitoring immunization coverage.

WUENIC
User reference

This guide uses an example country report to illustrate how the rules that govern the WUENIC process are applied to country-reported data in order to inform WUENIC coverage estimates using a four-phased approach. This guide may also be useful when reading country-specific [WUENIC time-series graphs online].

About the
WUENIC estimates

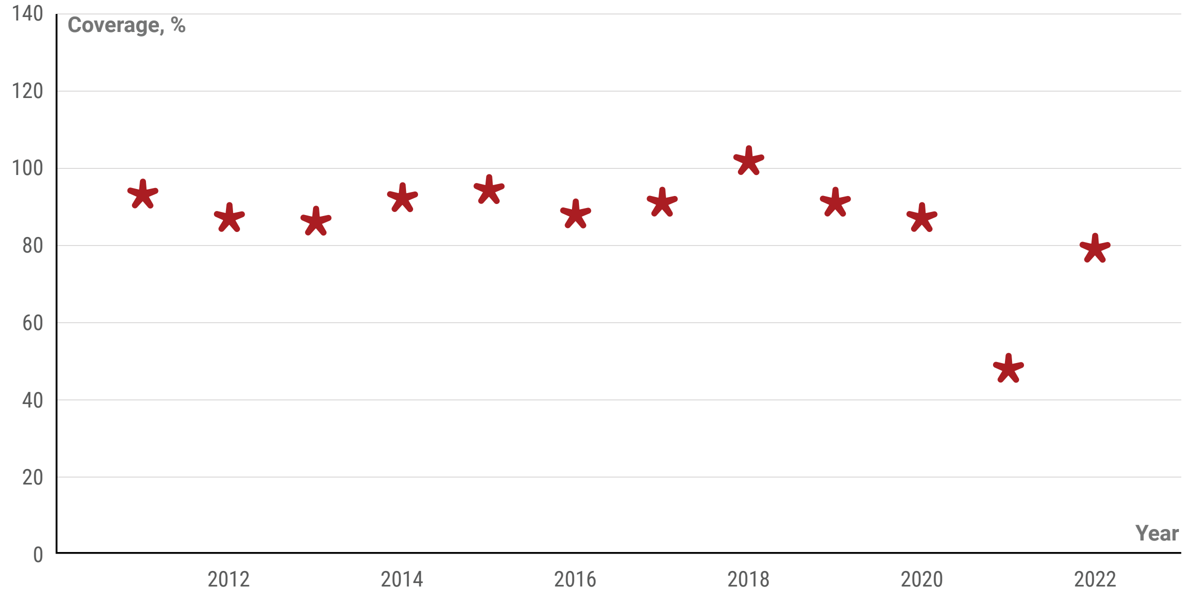
Since 2001, WHO and UNICEF have jointly produced and published annual estimates of national vaccination coverage (WUENIC) based on country-reported data. In addition to serving as a measure of an immunization programme's ability to effectively deliver immunization services and protect against vaccine-preventable diseases, WUENIC are used to monitor progress towards global and regional initiatives such as Immunization Agenda 2030 and Sustainable Development Goal (SDG) Indicator 3.b.1 related to monitoring the proportion of the target population covered by all vaccines included in a national programme.

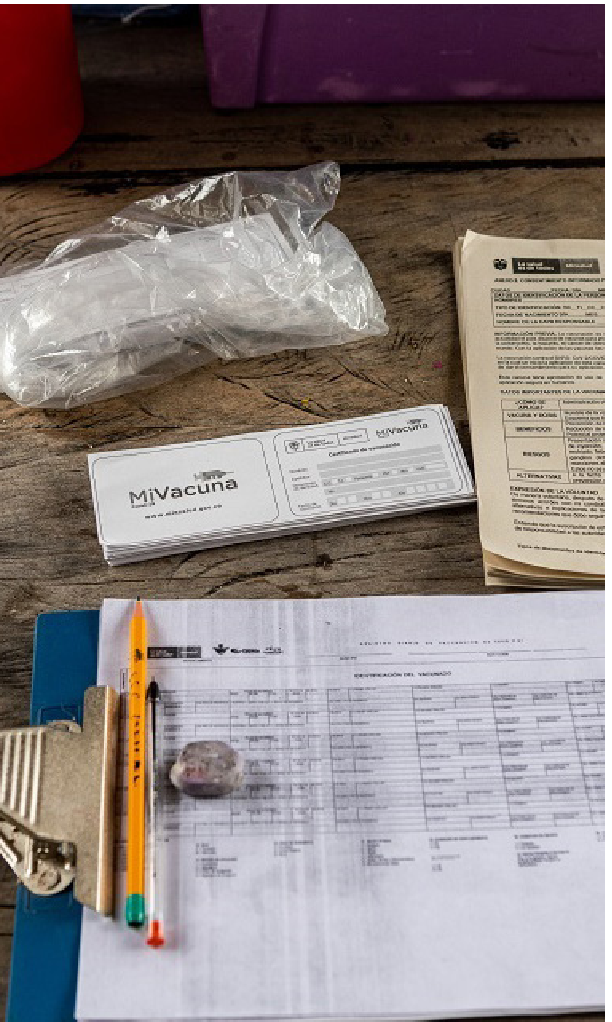
Phase 1

Administrative reports

WHO/UNICEF estimates of national infant routine vaccination coverage are based in part on coverage data derived from administrative data systems and reported through the [WHO/UNICEF Joint Reporting Form on Immunization (JRF)](#).

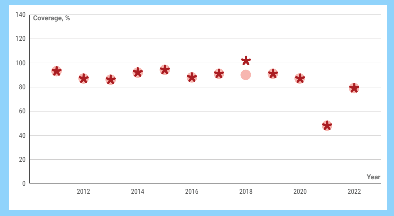
Administrative coverage data for an antigen are displayed as red stars (*) by year for the reporting period (x-axis). Coverage is presented as the percentage of a target population that has been vaccinated (y-axis). For example, administrative coverage for the third dose of DTP is calculated by dividing the number of children receiving the third dose of DTP vaccine by the number of children who survived to their first birthday.





WUENIC process

PHASE 2



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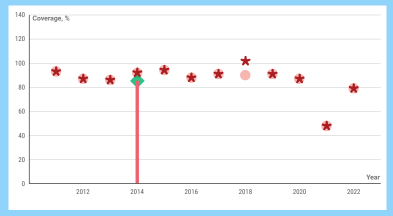

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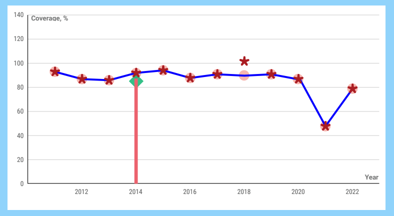
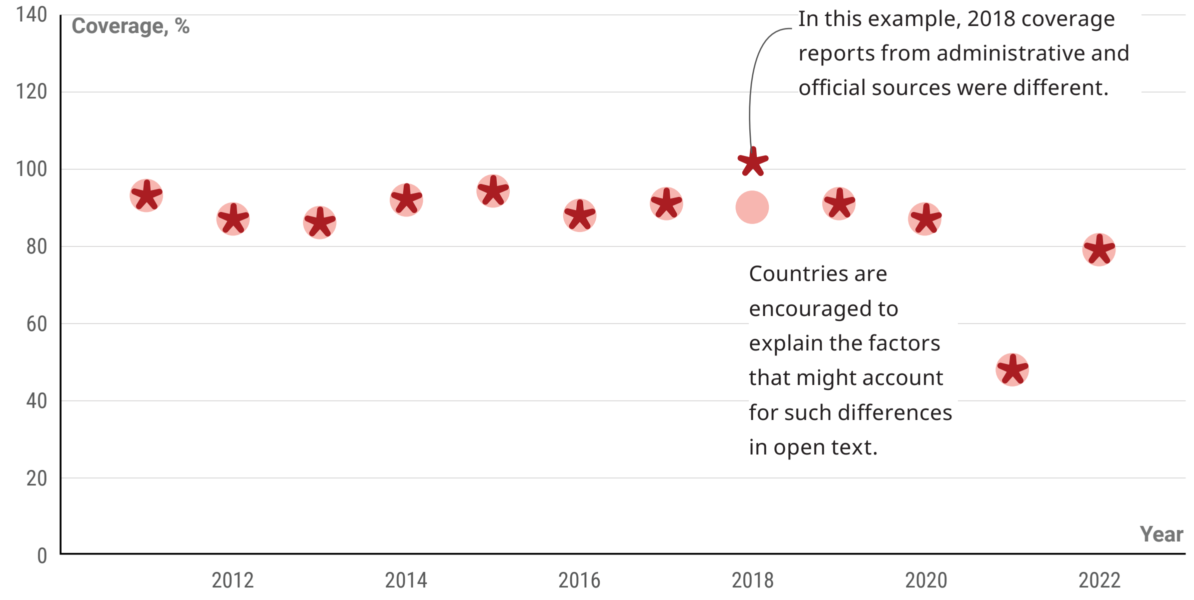


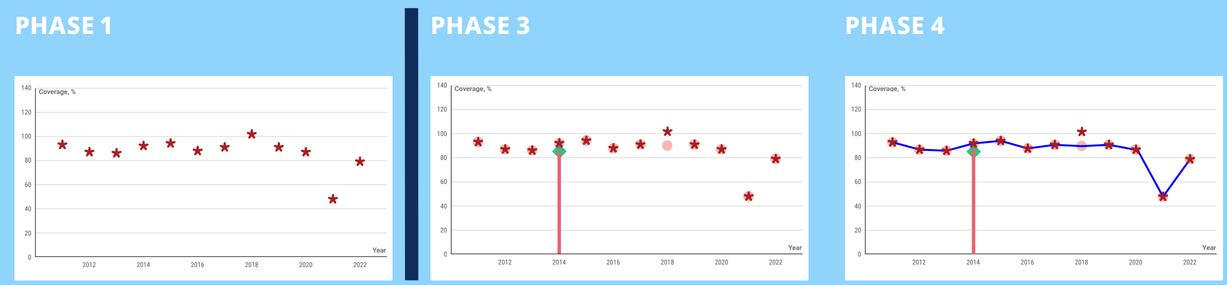
Phase 2
Official government estimates

National authorities are also asked to provide an official government estimate of vaccination coverage through the WHO/ UNICEF JRF. **Officially reported data are displayed as red circles** along with the administrative data (*) for the period.

Since administrative vaccination coverage can be biased or inaccurate, national authorities are given an opportunity to provide estimates of the most likely coverage — taking into account the administrative data, survey data (for example, in those instances where a programme does not have a well-functioning administrative monitoring system), and any other information on factors that may have affected vaccination coverage (for example, private or NGO sector delivery of vaccination, incomplete reporting or challenges with demographic data). **WHO and UNICEF recommend that any adjustments to officially reported coverage data be documented and reported with the estimates.**



WUENIC process



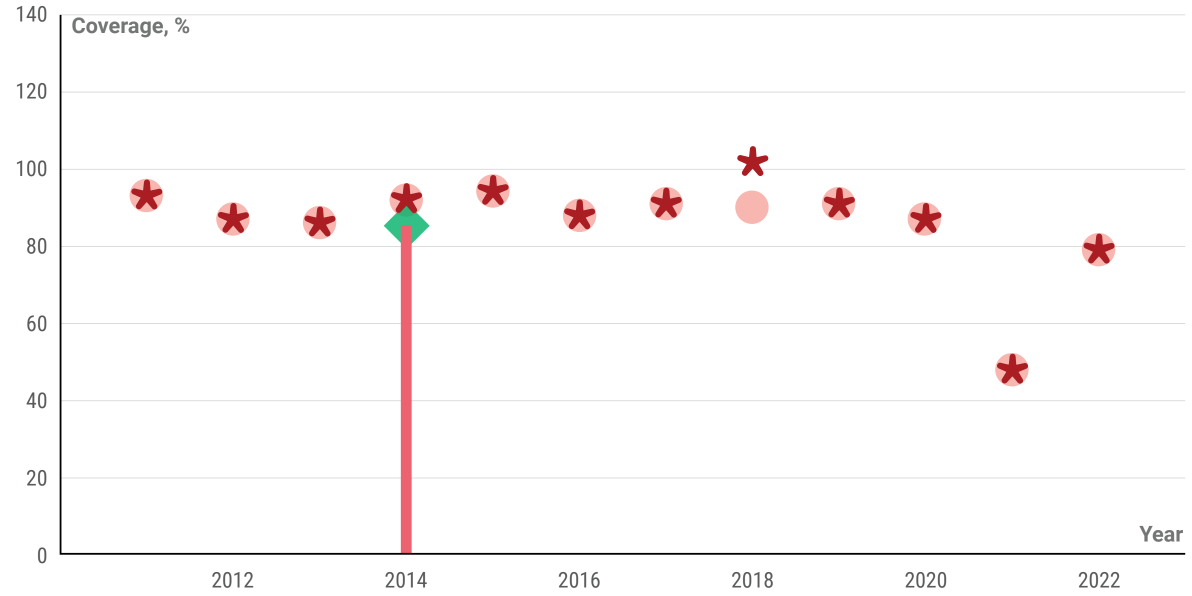
World Health Organization unicef

Phase 3

Survey reports

WHO and UNICEF also consider vaccination coverage estimates derived from high-quality surveys that can estimate coverage in the absence of accurate target population estimates (denominators) or from sectors not included in administrative reporting (for example, private or NGO contributions). **Survey results appear as vertical red bars, and survey values considered in producing WUENIC estimates are marked by a green diamond.** Survey bars without a green diamond are not considered (because of small sample size, inconsistencies in results across vaccine doses recommended at the same age or non-standard presentation of results). Survey results are described on the final page of the country report.

Survey results report on annual cohort(s) of children (for example, 12 to 23 months) and reflect the birth year of the cohort for infant vaccinations. Because surveys only provide information on previous birth cohorts, they may not provide timely information on programme interventions.



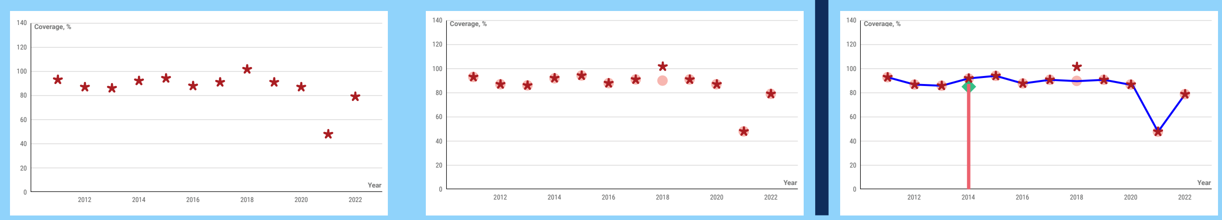
Coverage estimates can be biased, particularly for multi-dose antigens and in countries where home-based record distribution, retention and use are limited. Survey results marked with a green diamond for multi-dose antigens (for example, DTP3) may reflect a recall bias adjustment if survey information is available to compute an adjustment.

WUENIC process

PHASE 1

PHASE 2

PHASE 4



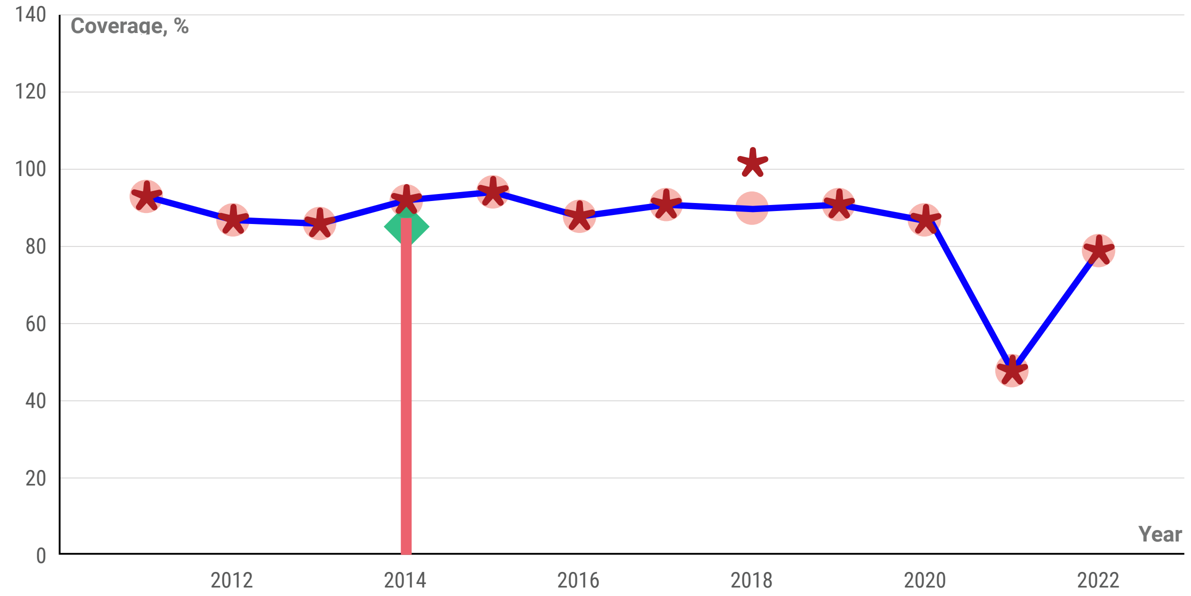
World Health Organization unicef

Phase 4

WUENIC estimated coverage

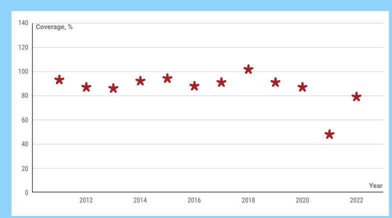
After considering all of the coverage data submitted by national authorities—administrative coverage, official government estimate and survey data—as well as contextual information (for example, stockouts, conflicts or natural disasters that might disrupt service delivery), WHO and UNICEF estimate the most likely coverage level for each country, vaccine and year. **The WUENIC estimate is displayed as a solid blue line.**

Additional information on the methods that guide the WHO and UNICEF estimation process can be found in the Resources section of this guide.



WUENIC
process

PHASE 1



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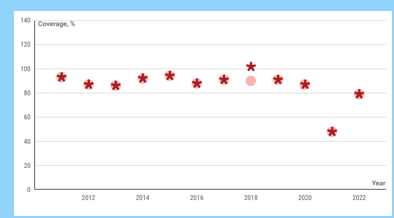

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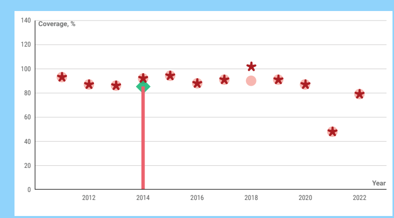
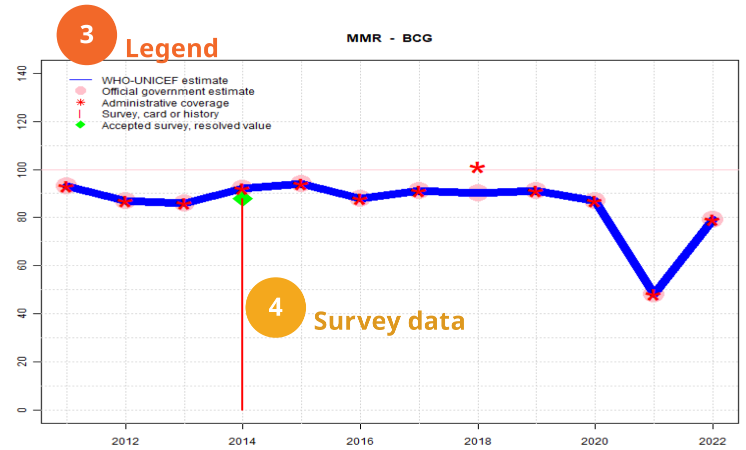


World Health Organization

unicef

Country 1 Zylandia - BCG 2 Antigen



	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Estimate	93	87	86	92	[illegible]	88	91	90	91	87	48	79
Estimate GoC	•	•	•	••	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	••	••
Official	93	87	86	92	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	48	79
Administrative	93	87	86	92	94	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	48	79
Survey	NA	NA	NA	88	NA	NA	NA	NA	NA	NA	NA	NA

5 Time series data

The WHO and UNICEF estimates of national immunization coverage (wuenic) are based on data and information that are of varying, and, in some instances, unknown quality. Beginning with the 2011 revision we describe the grade of confidence (GoC) we have in these estimates. As there is no underlying probability model upon which the estimates are based, we are unable to present classical measures of uncertainty, e.g., confidence intervals. Moreover, we have chosen not to make subjective estimates of plausibility/certainty ranges around the coverage. The GoC reflects the degree of empirical support upon which the estimates are based. It is not a judgment of the quality of data reported by national authorities.

●●● Estimate is supported by reported data [R+], coverage recalculated with an independent denominator from the World Population Prospects: 2022 revision from the UN Population Division (D+), and at least one supporting survey within 2 years [S[illegible]le well supported, the estimate still carries a risk of being wrong.

6 Grade of confidence

●● Estimate is supported by at least one data source; [R+], [S+], or [D+]; and no data source, [R-], [D-], or [S-], challenges the estimate.

● There are no directly supporting data; or data from at least one source; [R-], [D-], [S-]; challenge the estimate.

In all cases these estimates should be used with caution and should be assessed in light of the objective for which they are being used.

Description: 7 Text description

2022: Estimate informed by reported data. No nationally representative household survey within the last 5 years. WHO and UNICEF are aware of an ongoing 2023 Demographic and Health Survey and await the final results. Reported data suggest partial recovery from prior year disruption to services. GoC=R+ D+

2021: Estimate informed by reported data. WHO and UNICEF are aware of a EPI coverage evaluation survey that started in 2020 but was interrupted due to Covid-19. An independent assessment of reported coverage is still recommended. Routine immunization performance was low for all antigens due to disruptions caused by Covid-19 and instability. GoC=R+ D+

2020: Estimate informed by reported data. GoC=R+ D+

2019: Estimate informed by reported data. GoC=R+ D+

2018: Estimate informed by reported data. GoC=R+ D+

2017: Estimate informed by reported data. GoC=R+ D+

2016: Estimate informed by reported data. GoC=Assigned by working group. GoC revised for consistency with other vaccine doses.

2015: Estimate informed by reported data. GoC=Assigned by working group. In spite of what appear to be inconsistent reported number of children vaccinated and target population suggesting problems with the recording and monitoring system and/or incomplete reporting, the results of the 2015-16 Demographic and Health Survey support reported coverage levels. However, there is concern that less than half of the survey results are derived from documented evidence.

2014: Estimate informed by reported data supported by survey. Survey evidence of 88 percent based on 1 survey(s). Reported coverage levels for 2014 computed using target population data from preliminary 2014 census results. GoC=Assigned by working group. In spite of what appear to be inconsistent reported number of children vaccinated and target population suggesting problems with the recording and monitoring system and/or incomplete reporting, the results of the 2015-16 Demographic and Health Survey support reported coverage levels. However, there is concern that less than half of the survey results are derived from documented evidence.

2013: Estimate informed by reported data. Estimate challenged by: D-

2012: Estimate informed by reported data. Estimate challenged by: D-

2011: Estimate informed by reported data. Estimate challenged by: D-

8 Version

July 1, 2023; page 3 WHO and UNICEF estimates of national immunization coverage - next revision available July 15, 2024 data received as of June 26, 2023

How to read the WUENIC

Country Reports

1 **Country** name.

2 **Antigen** and, as applicable, dose combination for which estimates have been made.

3 **Legend** describing the four potential data series that may appear in the chart.

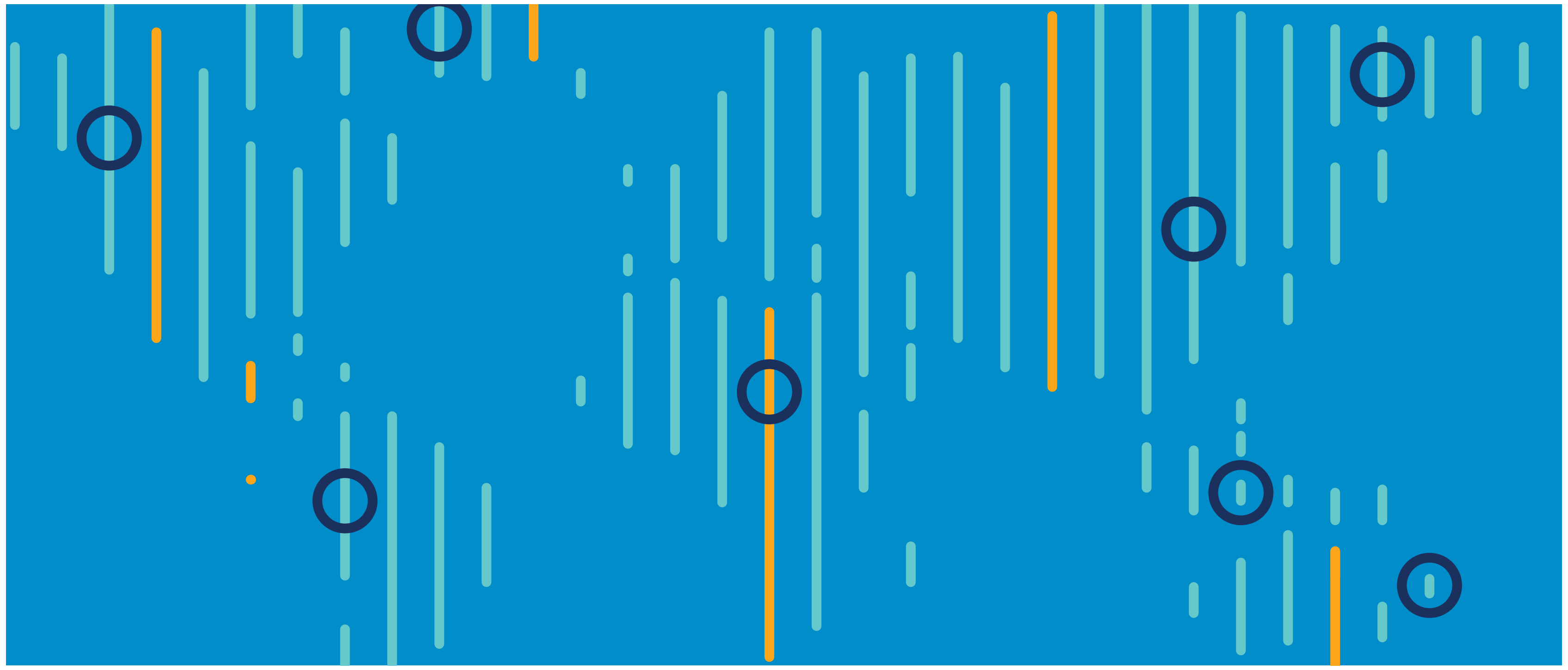
4 **Survey data** based on documented evidence (for example, home-based record card) or recall of vaccination history.

5 **Time series data** table of the values in the chart by data source, including the WHO/UNICEF estimate, official government estimate, administrative data, survey results, and the grade of confidence.

6 **Grade of confidence** classifications (*, **, ***) based on the level of supporting evidence available.

7 **Text description** that provides contextual information that informed the WUENIC estimate.

8 **Version** information, including data-as-of date and date of next revision. Please reference the version when citing WUENIC estimates.



> **"We must ensure that more children...are protected from vaccine-preventable diseases, allowing them to live happier, healthier lives."**
>
> **Tedros Adhanom Ghebreyesus**

Understanding WUENIC

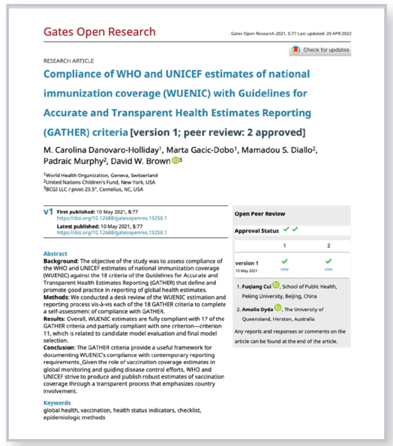
Resources

The following provide additional information on the WUENIC methods and process.


World Health Organization
Immunization Analysis and Insights
WHO/UNICEF estimates of national immunization coverage

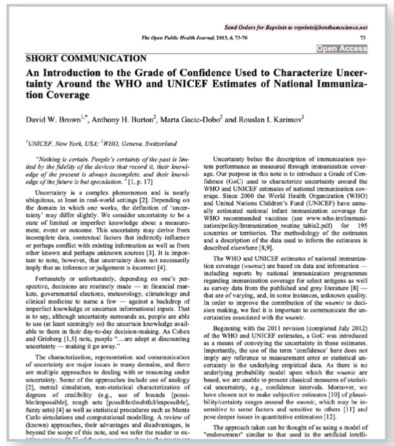
2024
WEBSITE & DATABASE

WHO/UNICEF estimates of national immunization coverage.


Gates Open Research
Compliance of WHO and UNICEF estimates of national immunization coverage (WUENIC) with Guidelines for Accurate and Transparent Health Estimates Reporting (GATHER) criteria [version 1; peer review: 2 approved]

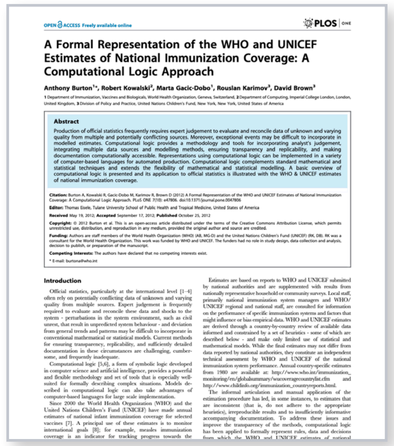
2021
GUIDELINES

Compliance of WHO and UNICEF estimates of national immunization coverage (WUENIC) with Guidelines for Accurate and Transparent Health Estimates Reporting (GATHER) criteria. *Danovaro-Holliday MC et al. Gates Open Res. 2021;5:77. doi: 10.12688/gatesopenres.13258.1*


SHORT COMMUNICATION
An Introduction to the Grade of Confidence Used to Characterize Uncertainty Around the WHO and UNICEF Estimates of National Immunization Coverage

2013
GRADE OF CONFIDENCE

An introduction to the Grade of Confidence used to characterize uncertainty around the WHO and UNICEF estimates of national immunization coverage. *Brown et al. Open Public Health J. 2013; 6:73-76.*


A Formal Representation of the WHO and UNICEF Estimates of National Immunization Coverage: A Computational Logic Approach

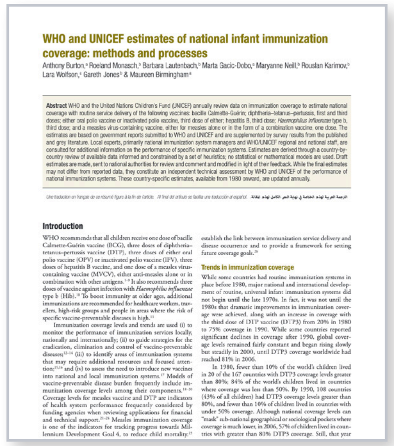
2012
METHODS & PROCESS

A formal representation of the WHO and UNICEF estimates of national immunization coverage: a computational logic approach. *Burton et al. PLoSOne. 2012;7(10):e47806.*


Manabu Okumura
Daisuke Bekki
Ken Satoh (Eds.)
LNAI 7258
New Frontiers in Artificial Intelligence
JSAI-isAI 2011 Workshops, LENLS, JURISIN, ALSIP, MiMI
Takamatsu, Japan, December 2011
Revised Selected Papers

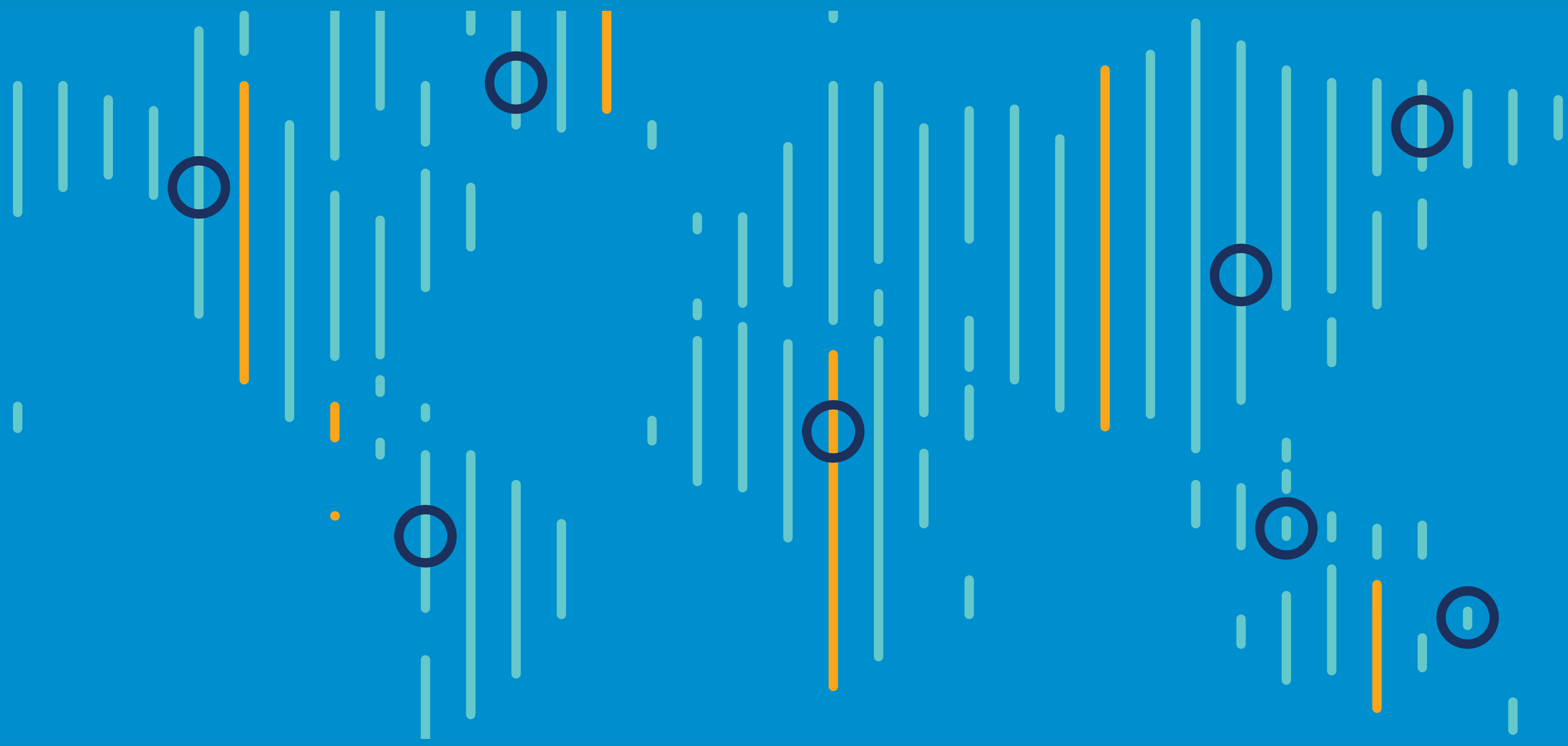
2011
METHODS & PROCESS

WUENIC – A Case Study in Rule-based Knowledge Representation and Reasoning. *Kowalski & Burton. In: Okumura M, Bekki D, Satoh K (eds) New Frontiers in Artificial Intelligence. JSAI-isAI 2011. Lecture Notes in Computer Science (7258).*


WHO and UNICEF estimates of national infant immunization coverage: methods and processes

2009
SEMINAL PUBLICATION

WHO and UNICEF estimates of national infant immunization coverage: methods and processes. *Burton et al. Bull World Health Organ. 2009;87(7):535-41.*



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